

Basal cell

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Authors: Honorary Associate Profes

General Practitioner and Cosmetic

Update peer reviewed by: Dr Andje
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Reviewing dermatologist: Dr Ian Cc

Edited by the DermNet content department.

Anthony Martin Fuentes,

bourne Hospital, Australia

What is basal cell carcinoma?

Basal cell carcinoma (BCC) is a common, locally invasive, [keratinocyte cancer](#) (also known as nonmelanoma cancer). It is the most common form of [skin cancer](#). BCC is also known as rodent ulcer and basalioma. Patients with BCC often develop multiple primary tumours over time.

BCC is also known as rodent ulcer and basalioma.

Who gets basal cell carcinoma?

Risk factors for BCC include:

Age and sex: BCCs are particularly prevalent in elderly males. However, they also affect females and younger adults

Previous BCC or other form of skin cancer ([squamous cell carcinoma](#), [melanoma](#))

Sun damage ([photoageing](#), [actinic keratoses](#))

Repeated prior episodes of [sunburn](#)

Fair skin, blue eyes and blond or red hair — note; BCC can also affect darker [skin types](#)

Previous cutaneous injury, [thermal burn](#), disease (eg [cutaneous lupus](#), [sebaceous naevus](#))

Inherited syndromes: BCC is a particular problem for families with basal cell naevus syndrome ([Gorlin syndrome](#)), [Bazex-Dupré-Christol syndrome](#), Rombo syndrome, Oley syndrome and [xeroderma pigmentosum](#)

Other risk factors include ionising radiation, exposure to [arsenic](#), immune suppression due to [disease](#) or [medicines](#), and use of some other medicines such as hydrochlorothiazide.

What causes basal cell carcinoma?

The cause of BCC is multifactorial.

Most often, there are DNA mutations in the patched (PTCH) tumour suppressor gene, part of the hedgehog signalling pathway.

These may be triggered by exposure to [ultraviolet radiation](#).

Various spontaneous and inherited [gene defects](#) predispose to BCC.

What are the clinical features of basal cell carcinoma?

BCC is a locally invasive skin tumour. The main characteristics are:

- Slowly growing plaque or nodule
- Skin coloured, pink or pigmented
- Varies in size from a few millimetres to several centimetres in diameter
- Spontaneous bleeding or ulceration.

BCC is very rarely a threat to life. A tiny proportion of BCCs grow rapidly, invade deeply, and/or metastasise to local lymph nodes.

See also: [Basal cell carcinoma in skin of colour](#)

Types of basal cell carcinoma

There are several distinct clinical types of BCC, and over 20 histological growth patterns of BCC.

Nodular BCC

- Most common type of facial BCC
- Shiny or pearly nodule with a smooth surface
- May have central depression or ulceration, so its edges appear rolled
- Blood vessels cross its surface
- Cystic variant is soft, with jelly-like contents
- Micronodular, microcystic and infiltrative types are potentially aggressive subtypes
- Also known as nodulocystic carcinoma



Nodular basal cell carcinoma



Nodular basal cell carcinoma



Nodular basal cell carcinoma



Nodular basal cell carcinoma

Superficial BCC

- Most common type in younger adults
- Most common type on the upper trunk and shoulders
- Slightly scaly, irregular plaque
- Thin, translucent rolled border
- Multiple microerosions



Basal cell carcinoma



Superficial basal cell carcinoma



Superficial basal cell carcinoma, face



Superficial basal cell carcinoma, back

Morphoeic BCC

- Usually found in mid-facial sites
- Waxy, scar-like plaque with indistinct borders

Wide and deep subclinical extension
May infiltrate cutaneous nerves (perineural spread)
Also known as morpheic, morphoeiform or sclerosing BCC



Morpheic basal cell carcinoma



Morpheic basal cell carcinoma



Morpheic basal cell carcinoma



Morpheic basal cell carcinoma

Basosquamous carcinoma

Mixed basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)
Infiltrative growth pattern
Potentially more aggressive than other forms of BCC
Also known as basosquamous carcinoma and mixed basal-squamous cell carcinoma



Basal cell carcinoma



Basosquamous cell carcinoma



Basisquamous cell carcinoma

See more images of basal cell carcinoma.

[BCC affecting the face images](#)

[BCC affecting the nose images](#)

[BCC affecting the eyelid images](#)

[BCC affecting the ear images](#)

[BCC affecting the trunk images](#)

What are the complications of basal cell carcinoma?

Recurrent BCC

Recurrence of BCC after initial treatment is not uncommon. Characteristics of recurrent BCC often include:

Incomplete excision or narrow margins at primary excision

Morphoeic, micronodular, and infiltrative subtypes

Location on head and neck.



After PDT



After superficial surgery



After fluorouracil



After imiquimod

Advanced BCC

Advanced BCCs are large, often neglected tumours.

They may be several centimetres in diameter

They may be deeply infiltrating into tissues below the skin

They are difficult or impossible to treat surgically

Metastatic BCC

Very rare

Primary tumour is often large, neglected or recurrent, located on head and neck, with aggressive subtype

May have had multiple prior treatments

May arise in-site exposed to ionising radiation

Can be fatal

How is basal cell carcinoma diagnosed?

BCC is diagnosed clinically by the presence of a slowly enlarging skin lesion with typical appearance.

BCC dermoscopy criteria

Menzies' algorithm (Menzies, 2000)

1. Absence of pigment network

2. Presence of ≥ 1 among:

Arborizing vessels (telangiectasia with tree-like branching)

Ulceration

Blue-gray ovoid nests

Blue-gray globules

Leaf-like areas

Spoke-wheel areas.

Additional criteria for non-pigmented, superficial BCC:

Shiny white to red structure-less areas
Short fine telangiectasia
Multiple small ulcerations (erosions).

Additional non-classical criteria:

Concentric structures
Blue-gray dots, in focus
Melanocytic-like criteria
Brown-black dots and globules
Blue-white veil-like structures
Non-classical vascular patterns (hairpin, glomerular, dotted, comma).

Structures under polarised light:

[Shiny white lines](#) — tend to be disorganised or arranged in parallel
Shiny white areas (blotches)
[Rosettes](#)
Rainbow pattern.

Additional recent criteria:

Stellate pattern — from the visible tumour edge; may be formed by white lines, vessels, or uneven skin surfaces
Negative maple leaf-like areas
Multiple aggregated yellow-white (MAY) globules.

The diagnosis and histological subtype is usually confirmed pathologically by a diagnostic biopsy or following [excision](#). See: [basal cell carcinoma pathology](#).

While histology remains the gold standard for BCC subtyping, non-invasive optical techniques like line-field confocal [optical coherence tomography \(LC-OCT\)](#) are increasingly utilized for BCC diagnosis and monitoring. LC-OCT has proven its effectiveness in real-world settings, enabling early detection, minimizing unnecessary biopsies, and providing valuable insights into disease progression, especially in cases where surgery is not indicated.

What is the treatment for primary basal cell carcinoma?

The treatment for a BCC depends on its type, size and location, the number to be treated, patient factors, and the preference or expertise of the doctor. Most BCCs are treated surgically. Long-term follow-up is recommended to check for new lesions and recurrence; the latter may be unnecessary if histology has reported wide clear margins.

Excision biopsy

[Excision](#) means the lesion is cut out and the skin stitched up.

Most appropriate treatment for nodular, infiltrative and morphoeic BCCs.
Should include 3 to 5 mm margin of normal skin around the tumour.
Very large lesions may require [flap](#) or [skin graft](#) to repair the defect.

A pathologist will report deep and lateral margins.

Further surgery is recommended for lesions that are incompletely excised.

Mohs micrographically controlled excision

Mohs micrographically controlled surgery involves examining carefully marked excised tissue under the microscope, layer by layer, to ensure complete excision.

Very high cure rates achieved by trained Mohs surgeons.

Used in high-risk areas of the face around eyes, lips, and nose.

Suitable for ill-defined, morphoeic, infiltrative and recurrent subtypes.

Large defects are repaired by **flap** or **skin graft**.

Superficial skin surgery

Superficial skin surgery comprises shave, **curettage**, and **electrocautery**. It is a rapid technique using **local anaesthesia** and does not require **sutures**.

Suitable for small, well-defined nodular or superficial BCCs.

Lesions are usually located on the trunk or limbs.

Wound is left open to heal by **secondary intention**.

Moist **wound dressings** lead to healing within a few weeks.

Eventual scar quality variable.

Cryotherapy

Cryotherapy is the treatment of a superficial skin lesion by freezing it, usually with liquid nitrogen.

Suitable for small superficial BCCs on covered areas of the trunk and limbs.

Best avoided for BCCs on head and neck, and distal to knees.

Double freeze-thaw technique.

Results in a blister that crusts over and heals within several weeks.

Leaves permanent white mark.

Photodynamic therapy

Photodynamic therapy (PDT) refers to a technique in which BCC is treated with a photosensitising chemical, and exposed to light several hours later.

Topical photosensitisers include aminolevulinic acid lotion and **methyl aminolevulinate cream**.

Suitable for low-risk small, superficial BCCs.

Best avoided if tumour in-site at high risk of recurrence.

Results in inflammatory reaction, maximal 3–4 days after procedure.

Treatment repeated 7 days after initial treatment.

Excellent cosmetic results.

Imiquimod cream

[Imiquimod](#) is an immune response modifier.

Best used for superficial BCCs less than 2 cm diameter.

Applied three to five times each week, for 6–16 weeks.

Results in a variable inflammatory reaction, maximal at three weeks.

Minimal scarring is usual.

Fluorouracil cream

[5-Fluorouracil cream](#) is a topical cytotoxic agent.

Used to treat small superficial basal cell carcinomas.

Requires prolonged course, eg twice daily for 6–12 weeks.

Causes inflammatory reaction.

Has high recurrence rates.

Radiotherapy

[Radiotherapy](#) or X-ray treatment can be used to treat primary BCCs or as adjunctive treatment if margins are incomplete.

Mainly used if surgery is not suitable.

Best avoided in young patients and in genetic conditions predisposing to skin cancer.

Best cosmetic results achieved using multiple fractions.

Typically, patient attends once-weekly for several weeks.

Causes inflammatory reaction followed by scar.

Risk of [radiodermatitis](#), late recurrence, and new tumours.

What is the treatment for advanced or metastatic basal cell carcinoma?

Locally advanced primary, recurrent or metastatic BCC requires multidisciplinary consultation. Often a combination of treatments is used.

[Surgery](#)

[Radiotherapy](#)

[Targeted therapy](#).

Targeted therapy refers to the hedgehog signalling pathway inhibitors, [vismodegib](#) and [sonidegib](#). These drugs have some important risks and side effects.

How can basal cell carcinoma be prevented?

The most important way to prevent BCC is to avoid [sunburn](#). This is especially important in childhood and early life. Fair-skinned individuals and those with a personal or family history of BCC should [protect their skin from sun exposure](#) daily, year-round and lifelong.

Stay indoors or under the shade in the middle of the day.

Wear [covering clothing](#).

Apply high protection factor SPF50+ broad-spectrum [sunscreens](#) generously to exposed skin if outdoors.

Avoid [indoor tanning](#).

Oral [nicotinamide](#) (vitamin B3) in a dose of 500 mg twice daily may reduce the number and severity of BCCs.

What is the outlook for basal cell carcinoma?

Most BCCs are cured by treatment. Cure is most likely if treatment is undertaken when the lesion is small.

About 50% of people with BCC develop a second one within 3 years of the first. They are also at increased risk of other [skin cancers](#), especially [melanoma](#). Regular [self-skin examinations](#) and long-term annual [skin checks](#) by an experienced health professional are recommended.